

ASSIGNMENT 4

Q1]

```
#include <stdio.h>
int multfact(int iNo)
{
    int mult = 1;
    int iFact = iNo/2;
    for (int iCnt = 1; iCnt <= iFact; iCnt++)
    {
        if (iNo % iCnt == 0)
        {
            mult *= iCnt;
        }
    }
    return mult;
}
```

```
int main()
{
    int iValue = 0;
    int iRet = 0;
    printf("Enter number: \n");
    scanf("%d", &iValue);
    iRet = multFact(iValue);
    printf("%d", iRet);
    return 0;
}
```

Time Complexity : $O(n)$


```
Q2] #include <stdio.h>
int factRev (int iNo)
{
    int ifact = iNo/2;
    printf("Factors of %d in decreasing order\n", iNo);
    for (int iCnt = ifact; iCnt >= 1; iCnt--)
    {
        if (iNo % iCnt == 0)
        {
            printf("%d ", iCnt);
        }
    }
}
```

```
int main()
{
    int iValue = 0;
    int iRet = 0;
    printf("Enter no: \n");
    scanf("%d", &iValue);
    factRev(iValue);
    return 0;
}
```

Time complexity: $O(n)$

Q3] #include <stdio.h>

int NonFact (int iNo)

```
{  
    printf("Non-factors of %d are: \n", iNo);  
    for( int iCnt = 1; iCnt <= iNo; iCnt++)  
    {  
        if (iNo % iCnt != 0)  
        {  
            printf("%d ", iCnt);  
        }  
    }  
}
```

int main()

```
{  
    int iValue = 0;  
    printf("Enter no: \n");  
    scanf("%d", &iValue);  
    NonFact(iValue);  
    return 0;  
}
```


Q4] #include <stdio.h>

```
int sumNonfact(int ino)
```

```
{
```

```
    int iSum = 0;
```

```
    for (int iCnt = 1; iCnt <= ino; iCnt++)
```

```
    {
```

```
        if (ino % iCnt != 0)
```

```
        {
```

```
            iSum += iCnt;
```

```
        }
```

```
    }
```

```
    return iSum;
```

```
}
```

```
int main()
```

```
{
```

```
    int iValue = 0;
```

```
    int iRet = 0;
```

```
    printf("Enter no \n");
```

```
    scanf("%d", &iValue);
```

```
    iRet = sumNonfact(iValue);
```

```
    printf("%d", iRet);
```

```
    return 0;
```

```
}
```

Time Complexity : $O(n)$

Q5] #include <stdio.h>

```
int factD, ff(int iNo)
```

```
{
    int iSumfact = 0;
    int iSumNonfact = 0;
```

```
for(int iCnt = 1; iCnt < iNo; iCnt++)
```

```
{
    if(iNo % iCnt == 0)
```

```
{
    iSumfact += iCnt;
```

```
}
```

```
else
```

```
{
```

```
    iSumNonfact += iCnt;
```

```
}
```

```
}
```

```
return iSumfact - iSumNonfact;
```

```
}
```

```
int main()
```

```
{
```

```
    int iValue = 0;
```

```
    int iRet = 0;
```

```
    printf("Enter no\n");
```

```
    scanf("%d", &iValue);
```

```
    iRet = factD, ff(iValue);
```

```
    printf("%d", iRet);
```

```
    return 0;
```

```
}
```

Time complexity: $O(n)$